

Recommended Course Sequence
Aerospace Engineering Option C – Avionics & Aerospace Systems (September Entry)
2026-2027 Academic Year

	SUMMER /1		FALL /2		WINTER /4	
YEAR 1			AERO 201 Intro to Flight & Aero Systems (4.00) The following course must be completed previously or concurrently: ENGR 213.		ELEC 273 Basic Circuit Analysis (3.50) The following course must be completed previously: ENGR 213; PHYS 205.	
			COEN 243 Programming Methodology I (3.50) The following course must be completed previously: MATH 204 (Cegep Mathematics 105).		ENCS 282 Technical Writing & Comm. (3.00) Students must have satisfied the requirements in Section 71.20.7 Writing Skills Requirement, by passing the Engineering Writing Test (EWT) or by passing ENCS 272 with a grade of C- or higher, prior to enrolling.	
			ENGR 213 Applied Ord. Differential Eq. (3.00) The following course must be completed previously or concurrently: MATH 204 (Cegep Mathematics 105). The following course must be completed previously: MATH 205 (Cegep Mathematics 203).		ENGR 243 Dynamics (3.00) The following courses must be completed previously: ENGR 213, ENGR 242.	
			ENGR 233 Applied Advanced Calculus (3.00) The following course must be completed previously: MATH 204 (Cegep Mathematics 105); MATH 205 (Cegep Mathematics 203).		ENGR 244 Mechanics of Materials (3.75) The following courses must be completed previously: ENGR 213; ENGR 242 or ENGR 245. The following courses must be completed previously or concurrently: ENGR 233.	
			ENGR 242 Statics (3.00) The following course must be completed previously or concurrently: ENGR 213. The following courses must be completed previously: PHYS 204; MATH 204.		ENGR 201 Professional Practice & Resp. (1.50) Prerequisites: none.	
YEAR 2			COEN 212 Digital Systems Design I (3.50) The following course must be completed previously: MATH 204 (Cegep Mathematics 105).		AERO 290 Introduction to Aircraft Design (3.00) The following course must be completed previously: AERO 201. The following course must be completed previously or concurrently: ENCS 282.	
			COEN 231 Introduction to Discrete Math. (3.00) The following course must be completed previously: MATH 204 (Cegep Mathematics 105).		AERO 371 Modelling and Control Systems (3.50) The following courses must be completed previously: PHYS 205; ENGR 213, ENGR 243. The following course must be completed previously or concurrently: ENGR 311 or ELEC 342 or ELEC 364.	
			ELEC 242 Continuous-Time Signals and Sys. (3.00) The following courses must be completed previously: ELEC 273; ENGR 213.		COEN 244 Programming Methodology II (3.00) The following course must be completed previously: COEN 243 or MECH 215 or MIAE 215.	
			ENGR 202 Sust. Dev. Enviro. Stewardship (1.50) Prerequisites: none.		ELEC 342 Discrete-Time Signals and Systems (3.50) The following course must be completed previously: ELEC 242 or ELEC 264.	
			ENGR 371 Probability & Stats in Eng. (3.00) The following courses must be completed previously: ENGR 213, ENGR 233.		AERO 253 Thermal Sciences for Aerospace (3.00) The following course must be completed previously: MATH 205; ENGR 213.	
YEAR 3			AERO 390 Aerospace Engr. Design Project (3.00) The following courses must be completed previously: AERO 290, AERO 371; ENCS 282.		COEN 311 Comp. Organization and Software (3.50) The following courses must be completed previously: COEN 212, COEN 243.	
			COEN 352 Data Structures and Algorithms (3.00) The following courses must be completed previously: COEN 231, COEN 244.		ELEC 483 Real-Time Comp. Control Systems (3.50) The following courses must be completed previously: AERO 371 or ELEC 372; ELEC 342 or ELEC 364.	
			ELEC 481 Linear Systems (3.50) The following course must be completed previously: AERO 371 or ELEC 372 or MECH 371.		ENGR 301 Engr. Manage. Principles Econ (3.00) Prerequisites: none.	
			ENGR 361 Fluid Mechanics I (3.00) The following courses must be completed previously: ENGR 213, ENGR 233, ENGR 251 or AERO 253.		ENGR 391 Numerical Methods in Engr. (3.00) The following courses must be completed previously: ENGR 213, ENGR 233; COMP 248 or COEN 243 or MECH 215 or MIAE 215 or BCEE 231.	
			MIAE 383 Applied Machine Learning MIAE (3.50) The following courses must be completed previously: ENGR 371; MIAE 215.		SOEN 341 Software Process and Practices (4.00) The following courses must be completed previously or concurrently: COMP 352 or COEN 352; ENCS 282.	
YEAR 4			AERO 417 Standards, Reg. and Certification (3.00) The following course must be completed previously: ENGR 201.		AERO 483 Integration of Avionics Systems (3.00) The following courses must be completed previously: AERO 482; ELEC 481.	
			AERO 482 Avionic Navigation Systems (3.00) The following courses must be completed previously: ENGR 371 or COMP 233; AERO 371 or ELEC 372 or MECH 370 or SOEN 385.		General Studies (3.00) (Undergraduate Calendar, Sec. 71.110)	
			ENGR 392 Impact of Technology on Society (3.00) The following courses must be completed previously: ENCS 282; ENGR 201, ENGR 202.			
			Technical Electives (UGRAD Calendar, Sec. 71.55) Review your advisement report for the number of credits required. Speak with your Undergraduate Program Assistant if you have any further questions			
			AERO 490 Capstone Aerospace Engineering Design Project (6.00) The following courses must be completed in advance: AERO 390; ENGR 301. Students must have completed 75 credits in the program prior to enrolling.			

DETAILED COURSE INFORMATION Aerospace - Option C 2026-27

COURSE	TITLE	CREDIT	PRE-REQUISITE	CO-REQUISITE	SUM 1	SUM 2	FALL	WIN
AERO 201	Introduction to Flight and Aerospace Systems	4.00	ENGR 213				X	X
AERO 253	Thermal Sciences for Aerospace Engineering	3.00	ENGR 213; MATH 205					X
AERO 290	Introduction to Aircraft Design	3.00	AERO 201	ENCS 282			X	X
AERO 371	Modelling and Control Systems	3.50	PHYS 205; ENGR 213, ENGR 243	ENGR 311 or ELEC 342 or ELEC 364			X	X
AERO 390	Aerospace Engineering Design Project	3.00	AERO 290, AERO 371; ENCS 282				X	X
AERO 417*	Standards, Regulations and Certification	3.00	ENGR 201		X*		X	
AERO 471	Aircraft Hydro-Mechanical and Fuel Systems	3.50	AERO 201. Or, permission of the Department.					X
AERO 472	Aircraft Pneumatic and Electrical Power Systems	3.50	AERO 201; ENGR 361		n/a	n/a	n/a	n/a
AERO 480	Flight Control Systems	3.50	AERO 371 or ELEC 372 or MECH 371 or SOEN 385				X	
AERO 482	Avionic Navigation Systems	3.00	ENGR 371 or COMP 233; AERO 371 or ELEC 372 or MECH 371 or SOEN 385				X	
AERO 483	Integration of Avionics Systems	3.00	AERO 482; ELEC 481					X
AERO 490	Capstone Aerospace Engineering Design Project	6.00	AERO 390; ENGR 301. Students must have completed 75 credits in the program.				X	
COEN 212	Digital Systems Design I	3.50	MATH 204 (Cegep Mathematics 105)			X	X	X
COEN 231	Introduction to Discrete Mathematics	3.00	MATH 204 (Cegep Mathematics 105)			X	X	X
COEN 243	Programming Methodology I	3.00	MATH 204 (Cegep Mathematics 105)		X		X	X
COEN 244	Programming Methodology II	3.00	COEN 243 or MECH 215 or MIAE 215		X		X	X
COEN 311	Computer Organization and Software	3.50	COEN 212, COEN 243		X		X	X
COEN 313	Digital Systems Design II	3.50	COEN 212, COEN 231				X	X
COEN 317	Microprocessor Systems	3.50	COEN 311 or COMP 228 or SOEN 228; COEN 313				X	X
COEN 320	Introduction to Real-Time Systems	3.50	COEN 346 or COMP 346				X	X
COEN 346	Operating Systems	3.50	COEN 311; COMP 352 or COEN 352				X	X
COEN 352	Data Structures and Algorithms	3.00	COEN 231, COEN 244		X		X	X
COEN 366	Communication Networks and Protocols	3.50	COEN 346				X	X
COEN 413	Hardware Functional Verification	3.50	COEN 313		n/a	n/a	n/a	n/a
COEN 421	Embedded Systems Design	4.00	COEN 317, COEN 320; SOEN 341					X
COEN 498	Topics in Computer Engineering	3.00	Permission of the Department is required.		n/a	n/a	n/a	n/a
ELEC 242	Continuous-Time Signals and Systems	3.00	ELEC 273; ENGR 213		X		X	X
ELEC 251	Fundamentals of Applied Electromagnetics	3.00	ELEC 273 or ENGR 273; ENGR 233				X	X
ELEC 273	Basic Circuit Analysis	3.50	ENGR 213; PHYS 205		X		X	X
ELEC 311	Electronics I	3.50	ELEC 273				X	X
ELEC 331	Fundamentals of Electrical Power Engineering	3.50	ELEC 251, ELEC 273				X	X
ELEC 342	Discrete-Time Signals and Systems	3.50	ELEC 242 or ELEC 264		X		X	X
ELEC 351	Electromagnetic Waves and Guiding Structures	3.50	ELEC 242, ELEC 251				X	X
ELEC 367	Introduction to Digital Communications	3.50	ELEC 342 or ELEC 364; ENGR 371				X	X
ELEC 433	Power Electronics	3.50	ELEC 311, ELEC 331				X	
ELEC 442	Advanced Signal Processing	3.50	ELEC 342 or ELEC 364; ENGR 371					X
ELEC 458	Techniques in Electromagnetic Compatibility	3.00	ELEC 351		X			
ELEC 464	Wireless Communications	3.00	ELEC 367				X	
ELEC 481	Linear Systems	3.50	AERO 371 or ELEC 372 or MECH 371				X	
ELEC 482	System Optimization	3.50	ENGR 391 or EMAT 391					X
ELEC 483	Real-Time Computer Control Systems	3.50	AERO 371 or ELEC 372; ELEC 342 or ELEC 364					X
ELEC 498	Topics in Electrical Engineering	3.00	Permission of the Department is required.		n/a	n/a	n/a	n/a
ENCS 282	Technical Writing and Communication	3.00	Passing the Engineering Writing Test (EWT) or ENCS 272 with a grade of C- or higher.		X	X	X	X
ENGR 201	Professional Practice and Responsibility	1.50			X		X	X
ENGR 202	Sustainable Development and Environmental Stewardship	1.50			X		X	X
ENGR 213	Applied Ordinary Differential Equations	3.00	MATH 205 (Cegep Mathematics 203)	MATH 204 (Cegep Mathematics 105)	X		X	X
ENGR 233	Applied Advanced Calculus	3.00	MATH 204 (Cegep Mathematics 105); MATH 205 (Cegep Mathematics 203)		X	X	X	X
ENGR 242	Statics	3.00	ENGR 213	PHYS 204; MATH 204	X		X	X
ENGR 243	Dynamics	3.00	ENGR 213, ENGR 242		X		X	X
ENGR 244	Mechanics of Materials	3.75	ENGR 213; ENGR 242 or ENGR 245	ENGR 233	X		X	X
ENGR 301	Engineering Management Principles and Economics	3.00			X	X	X	X
ENGR 361	Fluid Mechanics I	3.00	ENGR 213, ENGR 233, ENGR 251 or AERO 253		X		X	X
ENGR 371	Probability and Statistics in Engineering	3.00	ENGR 213, ENGR 233		X	X	X	X
ENGR 391	Numerical Methods in Engineering	3.00	ENGR 213, ENGR 233; COMP 248 or COEN 243 or MECH 215 or MIAE 215 or BCEE 231			X	X	X
ENGR 392	Impact of Technology on Society	3.00	ENCS 282; ENGR 201, ENGR 202		X	X	X	X
ENGR 411	Special Technical Report	1.00	ENCS 282. Permission of the Department is required.		X		X	X
ENGR 412	Honours Research Project	3.00	ENCS 282; 75cr in the BEng program, a CGPA of 3.00 or better. Permission of the Dept.		X		X	X
Gen. Ed.	General Education Elective	3.00	See section 71.7110 of the Undergraduate Calendar		X	X	X	X
MIAE 383	Machine Learning for MIAE	3.50	ENGR 371; MIAE 215 or COEN 243				X	X
SOEN 341	Software Process	4.00	COMP 352 or COEN 352; ENCS 282				X	X
SOEN 342	Software Requirements and Specifications	3.00	SOEN 341				X	X
SOEN 343	Software Architecture and Design	4.00	SOEN 341; SOEN 342				X	X

*AERO 417 reserved for AERO students in summer

Note: In the case of discrepancies between this and the current Undergraduate Calendar, please contact your Undergraduate Program Assistant for clarification.
This information was compiled April 2026.

Registration Essentials

How do I register for classes?

Student registration is accessible within the [Student Centre](#) or the [My CU Account](#) page of the Student Hub. Detailed information and how-to guides can be found here [Course registration - Concordia University](#).

If you need assistance for any particular reason, please email your [Undergraduate Program Assistant](#) and include your Student ID as well as detailed course information (i.e. term, lecture, lab and tutorial), as follows:

Information we need:	Where to find the information	Example:
Summer/Fall/Winter YEAR Lec ABCD 123-CC (2964) Tut ABCD 123-CCF (2969) Lab ABCD 123- CI-X (2966)	MIAE 215 Programming for Mechanical and Industrial Engineers LECT 7306 Seats: 62/90 LAB GI-X 7321 Seats: 21/24 TUT T TB 7308 Seats: 14/48 This is the course information that we need. ← Sample taken from the Visual Schedule Builder	Fall 2026 LEC MIAE 215 T (7306*) Tut MIAE 215 TB (7308*) Lab MIAE 215 GI-X (7321*) *Code is optional

Registration Regulation: The 'C -' Rule

All **200-level courses** within the program [including admission/ECP/MEP requirements] which are prerequisites for other courses must be completed with a C- or higher. A 200-level course in which a student has obtained a D+ or lower must be repeated before attempting a course for which it is a prerequisite (Undergraduate Calendar, [Section 71.10.4](#)). **This does not apply to 300 or 400 level courses.**

Registration Regulation: The '200-level before 400-level' Rule

Students must complete (with a posted final grade) all 200-level courses required for their program before registering for any 400-level courses (Undergraduate Calendar, [Section 71.10.4](#)).

Why can't I enroll for ENCS 282?

Before enrolling in ENCS 282, students must meet the Writing Skills Requirement (Undergraduate Calendar, [Section 71.20.7](#)) by either passing the Engineering Writing Test (EWT) or ENCS 272 with a grade of C or higher. **This is faculty policy, no exceptions.** More details about the EWT, including registration and contact info, can be found here: [Engineering Writing Test](#).

Where can I find the list of General Studies courses?

You can find the list of available courses here: [Section 71.110 Complementary Studies for Engineering and Computer Science Students](#). Please note the [Exclusion List](#), as these courses may not be taken to fulfill this requirement.

Do you need to re-sequence?

Please use [our website's](#) re-sequencing guide, planning tools, and [sequence template](#) to draft your course plan through to graduation. On page 2 of your recommended sequence, you'll find a table outlining all prerequisites, co-requisites, and course offerings by term—this will help with your planning. You can send a draft to your Program Assistant for review.

Registration: 3rd Time Repeat

A student may repeat a failed course only once. A student who fails a required course twice must request permission to take the course a third time. To submit a request, please go to: [GCS Student requests & forms](#)

How do I register for Capstone?

Online registration for Capstone opens June 1, after the annual assessment of GPA is performed by the Registrar's Office. For those taking prerequisite courses in either summer term, you can only register once the grades are available.

How do I know if I am on track for graduation?

The best way to assess what is required for your program is to look at your [Academic Requirements Report](#). The report outlines the courses and credits required for your program(s), including majors, minors, and electives. It helps you monitor completed and outstanding requirements, track total credits, and note any deficiencies or transfer credits. *This is a reference tool only—consult your program advisor for official guidance.*

All of the information above, and more, can be found on our [Quick Tips and FAQ \(MIAE\)](#)

For more information, or clarification, please do not hesitate to contact your [Undergraduate Program Assistant](#) or [Student Academic Services](#)